Supplementary materials on “Decoupling heterogeneity and correlation in positive multivariate continuous data: The Ge-Ga$_d$ framework with applications”

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This supplementary material contains three special cases of Ge-Ga$_d$ distribution family, and introduces their definitions, basic properties, statistical inference together with the corresponding mean regression models in detail. At the same time, we added some US algorithms we constructed and the related distributions we need to use. Some numerical experimental results not presented in the article will also be supplemented in this document.

S1. Three specific cases of Ge-Ga$_d$ distribution family

In §2.4, we assumed that the r.v. τ follows an arbitrary distribution defined on $\mathbb{R}_+$. In this section, we fix the distribution of τ (e.g., inverse gamma, log-symmetric generalized inverse Gaussian, and log-normal) and explore three specific cases of the Ge-Ga$_d$ distributions.

S1.1 Multivariate inverse gamma mixture of Ga distribution

A positive r.v. Y is said to follow the *inverse gamma* (IGa) distribution with shape parameter $\alpha(>0)$ and scale parameter $\beta(>0)$, denoted by $Y \sim \mathrm{IGa}(\alpha, \beta)$, if its *probability density function* (pdf) is

$$\mathrm{IGa}(y|\alpha,\beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} y^{-\alpha-1} \mathrm{e}^{-\beta/y}, \quad y > 0.$$

We have $E(Y) = \beta/(\alpha-1)$ (if $\alpha > 1$) and $\mathrm{Var}(Y) = \beta^2/[(\alpha-1)^2(\alpha-2)]$ (if $\alpha > 2$). It is easy to show that if $Y \sim \mathrm{IGa}(\alpha, \beta)$, then $c \cdot Y \sim \mathrm{IGa}(\alpha, c\beta)$ for any constant $c > 0$.

Let $\tau^* \sim \mathrm{IGa}(\lambda, \beta_\tau)$, then we have

$$\mu^*\tau^* \overset{\mathrm{d}}{=} \mu^* \cdot \mathrm{IGa}(\lambda, \beta_\tau) = \frac{\mu^*\beta_\tau}{\lambda-1} \cdot \mathrm{IGa}(\lambda, \lambda-1) \overset{\mathrm{d}}{=} \mu \cdot \tau,$$

where $\mu = \mu^*\beta_\tau/(\lambda-1)$ and $\tau \sim \mathrm{IGa}(\lambda, \lambda-1)$ with $E(\tau) = 1$, which can motivate the following definition.

S1.1.1 Definition and basic properties of the IGa-Ga$_d$ distribution

Definition S1.1 (The multivariate IGa-Ga distribution). Suppose that $\tau \sim \mathrm{IGa}(\lambda, \lambda-1)$ with $\lambda > 1$ and the MR $X_j|\tau \overset{\mathrm{ind}}{\sim} \mathrm{Ga}(\alpha, \mu_j\tau)$, then the random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^\top$ is said to follow the *d-dimensional inverse gamma mixture of Ga* (IGa-Ga$_d$) distribution, denoted by $X \sim \text{IGa-Ga}_d(\alpha, \boldsymbol{\mu}, \lambda)$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)^\top$. ‖

Setting $f_\tau(\tau|\boldsymbol{\lambda}) = \mathrm{IGa}(\tau|\lambda, \lambda-1)$, we obtain pdf of $\boldsymbol{X} \sim \text{IGa-Ga}_d(\alpha, \boldsymbol{\mu}, \lambda)$ as

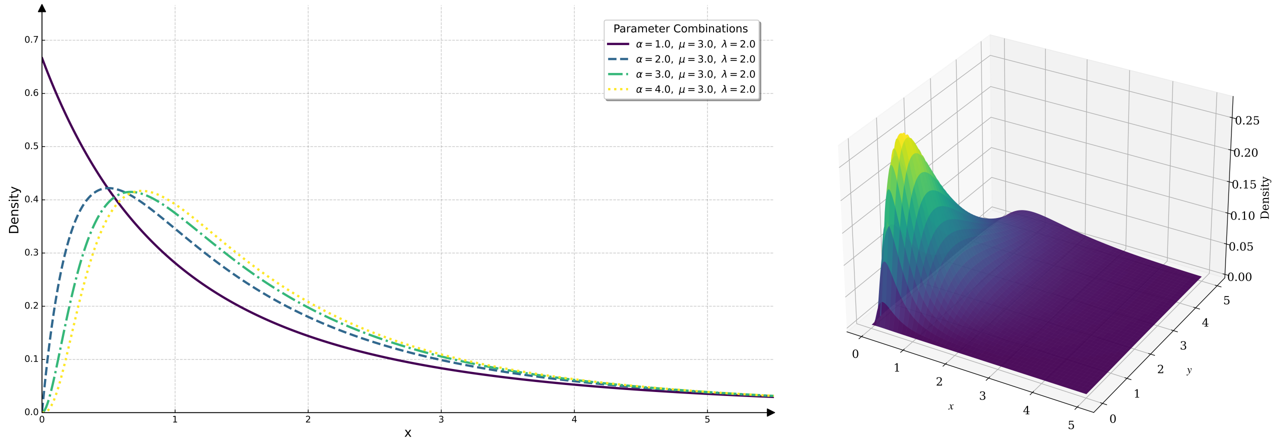
$$\begin{aligned}\text{IGa-Ga}_d(\boldsymbol{x}|\alpha, \boldsymbol{\mu}, \lambda) &= \frac{(\lambda-1)^\lambda}{\Gamma(\lambda)} \left[\prod_{j=1}^d \frac{\alpha^\alpha x_j^{\alpha-1}}{\mu_j^\alpha \Gamma(\alpha)}\right] \int_0^\infty h_{1*}(\tau|\boldsymbol{x}, \boldsymbol{\theta})\,\mathrm{d}\tau \\ &= \frac{\Gamma(\lambda+d\alpha)(\lambda-1)^\lambda}{\Gamma(\lambda)} \left[\prod_{j=1}^d \frac{\alpha^\alpha x_j^{\alpha-1}}{\mu_j^\alpha \Gamma(\alpha)}\right] \left(\sum_{j=1}^d \frac{\alpha x_j}{\mu_j} + \lambda - 1\right)^{-\lambda-d\alpha},\end{aligned}$$

where $\boldsymbol{\theta} \triangleq \{\alpha, \boldsymbol{\mu}, \lambda\}$, and

$$h_{1*}(\tau|\boldsymbol{x}, \boldsymbol{\theta}) = \tau^{-(\lambda+d\alpha)-1} \exp\left[-\left(\sum_{j=1}^d \frac{\alpha x_j}{\mu_j} + \lambda - 1\right) \cdot \frac{1}{\tau}\right], \quad \tau > 0. \tag{S1.1}$$

Next, for $j, j' = 1, \ldots, d$ and $j' \neq j$, we obtain the expectation, variance and covariance as

$$E(X_j) = \mu_j, \quad \mathrm{Var}(X_j) = \frac{\mu_j^2(\alpha+\lambda-1)}{\alpha(\lambda-2)} \quad \text{and} \quad \mathrm{Cov}(X_j, X_{j'}) = \frac{\mu_j\mu_{j'}}{\lambda-2}, \quad \text{if } \lambda > 2.$$



(a) IGa-Ga$_1$ (inverted Beta) distribution; (b) IGa-Ga$_2$ distribution;

Figure 1: The density functions of IGa-Ga$_d$ distributions.

Remark S1.1 (Alternative form of inverted Beta distribution). When dimension $d = 1$, we found that IGa-Ga$_d(\alpha, \boldsymbol{\mu}, \lambda)$ is reduced to an alternative form of inverted Beta distribution, whose pdf is

$$\mathrm{IBeta}(x|a, b, p) = \frac{p^b x^{a-1}}{\mathrm{Beta}(a, b)(p + x)^{a+b}}, \quad x > 0$$

with parameters $a = \alpha$, $b = \lambda$ and $p = \mu(\lambda - 1)/\alpha$, see (S2.2). In §S1.1.2, we will propose a new approach (i.e., the N-EM algorithm for the first time) to calculate the MLEs of IGa-Ga distribution (also known as inverted Beta distribution). ‖

S1.1.2 MLEs of parameters in the IGa-Ga$_d$ distribution

Let $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{iid}}{\sim}$ IGa-Ga$_d(\alpha, \boldsymbol{\mu}, \lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i\}_{i=1}^n$ denote the observed data, where $\boldsymbol{x}_i$ is the observation of $\boldsymbol{X}_i$, we obtain the log-likelihood function of $\boldsymbol{\theta}$ as

$$\begin{aligned} \ell_{1*}(\boldsymbol{\theta}|Y_{\text{obs}}) &= \sum_{i=1}^n \sum_{j=1}^d [\alpha \log \alpha + (\alpha - 1) \log x_{ij} - \log \Gamma(\alpha) - \alpha \log \mu_j] \\ &\quad + n[\lambda \log(\lambda - 1) - \log \Gamma(\lambda)] + \sum_{i=1}^n \log \int_0^\infty h_{1*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}\tau, \end{aligned}$$

where $h_{1*}(\cdot|\boldsymbol{x}_i, \boldsymbol{\theta})$ is defined by (S1.1). By defining the ndf

$$g_{1*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}) \triangleq \frac{h_{1*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})}{\int_0^\infty h_{1*}(s|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}s} = \mathrm{IGa}\left(\tau \middle| \lambda + d\alpha,\ \textstyle\sum_{j=1}^d \alpha x_{ij} \mu_j^{-1} + \lambda - 1\right),$$

we can construct the surrogate function as

$$
\begin{aligned}
Q_{1*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) &= c_{1*}^{(t)} + n\left[\lambda\log(\lambda-1) - \log\Gamma(\lambda) - \left(\bar{b}_{1*}^{(t)} + \bar{d}_{1*}^{(t)}\right)\lambda\right] \\
&\quad + n\sum_{j=1}^{d}[\alpha\log\alpha - \log\Gamma(\alpha) - \alpha\log\mu_j] - nd\bar{d}_{1*}^{(t)}\alpha \\
&\quad + \sum_{i=1}^{n}\sum_{j=1}^{d}\left(\alpha\log x_{ij} - b_{i,1*}^{(t)}x_{ij}\alpha\mu_j^{-1}\right),
\end{aligned}
$$

where $c_{1*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$ and for $1,\ldots,n$,

$$
\begin{aligned}
b_{i,1*}^{(t)} &= \int_0^\infty \frac{1}{\tau}\cdot g_{1*}(\tau|\boldsymbol{x}_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \qquad \bar{b}_{1*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n} b_{i,1*}^{(t)}, \\
d_{i,1*}^{(t)} &= \int_0^\infty \log\tau\cdot g_{1*}(\tau|\boldsymbol{x}_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad \bar{d}_{1*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n} d_{i,1*}^{(t)}.
\end{aligned}
$$

We obtain the following iterative expressions:

$$
\begin{aligned}
\alpha^{(t+1)} &= \mathrm{sol}\left\{C_{1*}^{(t)} + \log\alpha - \psi(\alpha) = 0\right\}, \\
\mu_j^{(t+1)} &= \frac{1}{n}\sum_{i=1}^{n} b_{i,1*}^{(t)}x_{ij}, \quad j = 1,\ldots,d, \\
\lambda^{(t+1)} &= \mathrm{sol}\left\{1 - \bar{b}_{1*}^{(t)} - \bar{d}_{1*}^{(t)} + \log(\lambda-1) + \frac{1}{\lambda-1} - \psi(\lambda) = 0\right\}, \qquad \text{(S1.2)}
\end{aligned}
$$

where the constant

$$
C_{1*}^{(t)} \triangleq 1 - \bar{d}_{1*}^{(t)} - \frac{1}{d}\sum_{j=1}^{d}\log\mu_j + \frac{1}{nd}\sum_{i=1}^{n}\sum_{j=1}^{d}\left(\log x_{ij} - b_{i,1*}^{(t)}x_{ij}\mu_j^{-1}\right),
$$

and the solution in (S1.2) can be solved by the US algorithm as shown in §S2.1.

S1.1.3 IGa-Ga$_d$ mean regression model

Assume that $\{\boldsymbol{X}_i\}_{i=1}^{n} \overset{\text{ind}}{\sim} \text{IGa-Ga}_d(\alpha,\boldsymbol{\mu}_i,\lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i,\boldsymbol{z}_{(i)}\}_{i=1}^{n}$ denote the observed data, we consider the following mean regression model

$$
\log E(X_{ij}) = \log\mu_{ij} = \boldsymbol{z}_{(i)}^{\top}\boldsymbol{\beta}_j \quad \text{or} \quad \mu_{ij} = \exp(\boldsymbol{z}_{(i)}^{\top}\boldsymbol{\beta}_j), \quad i = 1,\ldots,n,\ j = 1,\ldots,d,
$$

where $\boldsymbol{z}_{(i)} = (z_{i1}, \ldots, z_{iq})^\top$ is a q-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta}_j = (\beta_{1j}, \ldots, \beta_{qj})^\top$ is the regression coefficients with $q \times d + 2 \leqslant n$. The log-likelihood function of parameter vector $\boldsymbol{\gamma} = \{\alpha, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_d, \lambda\}$ is

$$\begin{aligned} \ell_1(\boldsymbol{\gamma}|Y_{\rm obs}) &= nd[\alpha\log\alpha - \log\Gamma(\alpha)] + \sum_{i=1}^n\sum_{j=1}^d [(\alpha-1)\log x_{ij} - \alpha\log\mu_{ij}] \\ &\quad + n[\lambda\log(\lambda-1) - \log\Gamma(\lambda)] + \sum_{i=1}^n \log\int_0^\infty h_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}\tau, \end{aligned}$$

where $h_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) = \tau^{-\lambda-d\alpha-1}\exp\{-[\sum_{j=1}^d(\alpha x_{ij})/\mu_{ij} + \lambda - 1]/\tau\}$. By defining the ndf

$$g_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) \triangleq \frac{h_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})}{\int_0^\infty h_1(s|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}s} = \mathrm{IGa}\left(\tau\middle|\lambda + d\alpha,\ \textstyle\sum_{j=1}^d \alpha x_{ij}\mu_{ij}^{-1} + \lambda - 1\right),$$

we construct the surrogate function as

$$\begin{aligned} Q_1(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)}) &= c_1^{(t)} + n\left[\lambda\log(\lambda-1) - \log\Gamma(\lambda) - \left(\bar{b}_1^{(t)} + \bar{d}_1^{(t)}\right)\lambda\right] - nd\bar{d}_1^{(t)}\alpha \\ &\quad + nd\left[\alpha\log\alpha - \log\Gamma(\alpha)\right] + \sum_{i=1}^n\sum_{j=1}^d\left(\alpha\log x_{ij} - \alpha\log\mu_{ij} - b_{i,1}^{(t)}x_{ij}\alpha\mu_{ij}^{-1}\right), \end{aligned}$$

where $c_1^{(t)}$ is a constant free from $\boldsymbol{\gamma}$, and for $i = 1, \ldots, n$,

$$\begin{aligned} b_{i,1}^{(t)} &= \int_0^\infty \frac{1}{\tau}\cdot g_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \qquad \bar{b}_1^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n b_{i,1}^{(t)}, \\ d_{i,1}^{(t)} &= \int_0^\infty \log\tau\cdot g_1(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \quad \bar{d}_1^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n d_{i,1}^{(t)}. \end{aligned}$$

Finally, we obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\gamma}}$ as

$$\begin{aligned} \alpha^{(t+1)} &= \mathrm{sol}\left\{C_1^{(t)} + \log\alpha - \psi(\alpha) = 0\right\}, \\ \boldsymbol{\beta}_j^{(t+1)} &= \left(\mathbf{Z}^\top\mathbf{W}_1^{(t)}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}_1^{(t)}\boldsymbol{r}_{j,1}^{(t)}, \quad j = 1, \ldots, d, \\ \lambda^{(t+1)} &= \mathrm{sol}\left\{1 - \bar{b}_1^{(t)} - \bar{d}_1^{(t)} + \log(\lambda-1) + \frac{1}{\lambda-1} - \psi(\lambda) = 0\right\}, \end{aligned}$$

where the constant

$$C_1^{(t)} \triangleq 1 - \bar{d}_1^{(t)} + \frac{1}{nd}\sum_{i=1}^n\sum_{j=1}^d\left(\log x_{ij} - \log\mu_{ij} - b_{i,1}^{(t)}x_{ij}\mu_{ij}^{-1}\right),$$

$\mathbf{Z} = (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})^\top$ is the matrix of covariates, $\mathbf{W}_1^{(t)} = \mathrm{diag}\{b_{1,1}^{(t)}, \ldots, b_{n,1}^{(t)}\}$ is the weighted matrix, $\boldsymbol{r}_{j,1}^{(t)} = (r_{1j,1}^{(t)}, \ldots, r_{nj,1}^{(t)})^\top$ with $r_{ij,1}^{(t)} = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j^{(t)} + (x_{ij}/\mu_{ij}^{(t)} - 1/b_{i,1}^{(t)})$ is the response vector for $i = 1, \ldots, n$.

S1.2 Multivariate log-symmetric GIGau mixture of Ga distribution

A positive r.v. is said to follow the *generalized inverse Gaussian* (GIGau) distribution (see more details in §S2.2) with scale parameters $a(>0)$, $b(>0)$ and shape parameter $p(\in \mathbb{R})$, if its pdf is

$$\mathrm{GIGau}(y|a,b,p) = \frac{(a/b)^{p/2}}{2K_p(\sqrt{ab})} y^{p-1} \exp\left[-\frac{1}{2}\left(ay + \frac{b}{y}\right)\right], \quad y > 0,$$

where $K_p(z)$ being the modified Bessel function of the second kind defined as (S2.1). If $Y \sim \mathrm{GIGau}(a,b,p)$, we obtain the expectation and variance

$$E(Y) = \sqrt{\frac{b}{a}} \frac{K_{p+1}(\sqrt{ab})}{K_p(\sqrt{ab})} \quad \text{and} \quad \mathrm{Var}(Y) = \frac{b}{a} \frac{K_{p+2}(\sqrt{ab})}{K_p(\sqrt{ab})} - \left[\sqrt{\frac{b}{a}} \frac{K_{p+1}(\sqrt{ab})}{K_p(\sqrt{ab})}\right]^2.$$

And for any constant $c > 0$, we easily show that $c \cdot Y \sim \mathrm{GIGau}(a/c, bc, p)$. Therefore, suppose that $\tau^* \sim \mathrm{GIGau}(a_\tau, b_\tau, p)$, we have

$$\mu^* \cdot \tau^* \overset{\mathrm{d}}{=} \mu^* \cdot \mathrm{GIGau}(a_\tau, b_\tau, p) = \mu^* \sqrt{\frac{b_\tau}{a_\tau}} \frac{K_{p+1}(\lambda)}{K_p(\lambda)} \cdot \mathrm{GIGau}(a_p(\lambda), b_p(\lambda), p) \overset{\mathrm{d}}{=} \mu \cdot \tau,$$

where $\mu = \mu^* \sqrt{b_\tau/a_\tau} K_{p+1}(\lambda)/K_p(\lambda)$ with $\lambda = \sqrt{a_\tau b_\tau}$,

$$a_p(\lambda) = \frac{\lambda K_{p+1}(\lambda)}{K_p(\lambda)} \quad \text{and} \quad b_p(\lambda) = \frac{\lambda K_p(\lambda)}{K_{p+1}(\lambda)}.$$

Then we obtain the mixture variable $\tau \sim \mathrm{GIGau}(a_p(\lambda), b_p(\lambda), p)$ with $E(\tau) = 1$. If we set the shape parameter p of the GIGau distribution to be zero, we will get the log-symmetric GIGau (LS-GIGau) distribution, which satisfies the symmetry on the logarithmic scale, meaning that mixture variable τ remains neutral on the multiplicative scale, which can motivate the following definition.

S1.2.1 Definition and basic properties of the LSGIGau-Ga$_d$ distribution

Definition S1.2 (The multivariate LSGIGau-Ga distribution). Let $\tau \sim \mathrm{GIGau}(a_0(\lambda), b_0(\lambda), 0)$ and $X_j|\tau \overset{\mathrm{ind}}{\sim} \mathrm{Ga}(\alpha, \mu_j \tau)$, then the random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^\top$ is said to follow the *d-dimensional log-symmetric generalized inverse Gaussian mixture of Ga* (LSGIGau-Ga$_d$) distribution, denoted by $X \sim$ LSGIGau-Ga$_d(\alpha, \boldsymbol{\mu}, \lambda)$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)^\top$. $\|$

By setting the mixture density $f_\tau(\tau|\boldsymbol{\lambda}) = \text{GIGau}(\tau|a_0(\lambda), b_0(\lambda), 0)$, we obtain pdf of $\boldsymbol{X} \sim$ LSGIGau-Ga$_d(\alpha, \boldsymbol{\mu}, \lambda)$ as

$$
\begin{aligned}
\text{LSGIGau-Ga}_d(\boldsymbol{x}|\alpha, \boldsymbol{\mu}, \lambda) &= \frac{1}{2K_0(\lambda)} \left[\prod_{j=1}^{d} \frac{\alpha^\alpha x_j^{\alpha-1}}{\Gamma(\alpha)\mu_j^\alpha}\right] \int_0^\infty h_{2*}(\tau|\boldsymbol{x}, \boldsymbol{\theta})\, \mathrm{d}\tau \\
&= \left[\prod_{j=1}^{d} \frac{\alpha^\alpha x_j^{\alpha-1}}{\Gamma(\alpha)\mu_j^\alpha}\right] \frac{K_{d\alpha}\left(\sqrt{a_0(\lambda)\left[b_0(\lambda) + 2\sum_{j=1}^d \alpha x_j/\mu_j\right]}\right)}{K_0(\lambda)} \\
&\quad \times \left[\frac{a_0(\lambda)}{b_0(\lambda) + 2\sum_{j=1}^d \alpha x_j/\mu_j}\right]^{d\alpha/2},
\end{aligned}
$$

where $\boldsymbol{\theta} \triangleq \{\alpha, \boldsymbol{\mu}, \lambda\}$, and the function $h_{2*}(\tau|\boldsymbol{x}, \boldsymbol{\theta})$ is defined as

$$
h_{2*}(\tau|\boldsymbol{x}, \boldsymbol{\theta}) = \tau^{-d\alpha-1} \exp\left\{-\frac{1}{2}\left[a_0(\lambda)\tau + \left(b_0(\lambda) + 2\sum_{j=1}^{d} \frac{\alpha x_j}{\mu_j}\right) \cdot \frac{1}{\tau}\right]\right\}, \quad \tau > 0. \quad \text{(S1.3)}
$$

For $j, j' = 1, \ldots, d$ and $j' \neq j$, we obtain the expectation, variance and covariance as

$$
\begin{aligned}
&E(X_j) = \mu_j, \quad \text{Var}(X_j) = \mu_j^2 \left[\left(1 + \frac{1}{\alpha}\right) \frac{K_2(\lambda)K_0(\lambda)}{K_1^2(\lambda)} - 1\right] \quad \text{and} \\
&\text{Cov}(X_j, X_{j'}) = \mu_j \mu_{j'} \left[\frac{K_2(\lambda)K_0(\lambda)}{K_1^2(\lambda)} - 1\right].
\end{aligned}
$$

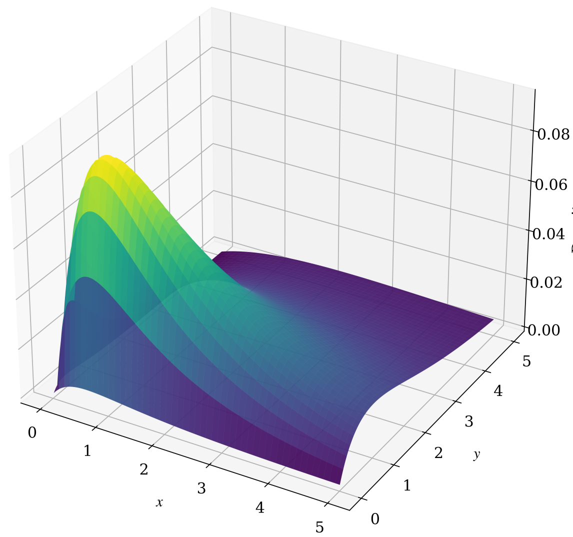


Figure 2: The density function of LSGIGau-Ga$_2$ distribution with $\alpha = 2$, $\boldsymbol{\mu} = (5, 3)^{\top}$, $\lambda = 2$.

S1.2.2 MLEs of parameters in the LSGIGau-Ga$_d$ distribution

Let $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{iid}}{\sim}$ LSGIGau-Ga$_d(\alpha, \boldsymbol{\mu}, \lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i\}_{i=1}^n$ denote the observed data, where $\boldsymbol{x}_i$ is the observation of $\boldsymbol{X}_i$, then, the log-likelihood function of $\boldsymbol{\theta}$ is given by

$$\begin{aligned}
\ell_{2*}(\boldsymbol{\theta}|Y_{\text{obs}}) \;&=\; \sum_{i=1}^n \sum_{j=1}^d [\alpha \log \alpha + (\alpha - 1) \log x_{ij} - \log \Gamma(\alpha) - \alpha \log \mu_j] \\
&\quad - n \log 2 - n \log K_0(\lambda) + \sum_{i=1}^n \log \int_0^\infty h_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}\tau,
\end{aligned}$$

where $h_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})$ is defined by (S1.3). By defining the ndf

$$g_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}) \triangleq \frac{h_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})}{\int_0^\infty h_{2*}(s|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}s} = \text{GIGau}\left(\tau \middle| a_0(\lambda), b_0(\lambda) + 2\textstyle\sum_{j=1}^d \alpha x_{ij} \mu_j^{-1}, -d\alpha\right),$$

we can construct the surrogate function as

$$\begin{aligned}
Q_{2*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) \;&=\; c_{2*}^{(t)} + n \sum_{j=1}^d [\alpha \log \alpha - \log \Gamma(\alpha) - \alpha \log \mu_j] - nd\bar{d}_{2*}^{(t)} \alpha \\
&\quad + \sum_{i=1}^n \sum_{j=1}^d (\alpha \log x_{ij} - b_{i,2*}^{(t)} x_{ij} \alpha \mu_j^{-1}) \\
&\quad - n \log K_0(\lambda) - \frac{n\bar{a}_{2*}^{(t)}}{2} a_0(\lambda) - \frac{n\bar{b}_{2*}^{(t)}}{2} b_0(\lambda),
\end{aligned}$$

where $c_{2*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$ and for $1, \ldots, n$,

$$
\begin{aligned}
a_{i,2*}^{(t)} &= \int_0^\infty \tau \cdot g_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \qquad \bar{a}_{2*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n a_{i,2*}^{(t)},\\
b_{i,2*}^{(t)} &= \int_0^\infty \frac{1}{\tau} \cdot g_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \qquad \bar{b}_{2*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n b_{i,2*}^{(t)},\\
d_{i,2*}^{(t)} &= \int_0^\infty \log\tau \cdot g_{2*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad \bar{d}_{2*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n d_{i,2*}^{(t)}.
\end{aligned}
$$

Finally, we obtain the following iterative expressions:

$$
\begin{aligned}
\alpha^{(t+1)} &= \mathrm{sol}\left\{C_{2*}^{(t)} + \log\alpha - \psi(\alpha) = 0\right\},\\
\mu_j^{(t+1)} &= \frac{1}{n}\sum_{i=1}^n b_{i,2*}^{(t)} x_{ij}, \quad j = 1, \ldots, d,\\
\lambda^{(t+1)} &= \arg\max_{\lambda>0} -\frac{n}{2}\left\{2\log K_0(\lambda) + \left[\frac{\bar{a}_{2*}^{(t)} K_1(\lambda)}{K_0(\lambda)} + \frac{\bar{b}_{2*}^{(t)} K_0(\lambda)}{K_1(\lambda)}\right]\lambda\right\},
\end{aligned}
$$

where the constant

$$
C_{2*}^{(t)} \triangleq 1 - \bar{d}_{2*}^{(t)} - \frac{1}{d}\sum_{j=1}^d \log\mu_j + \frac{1}{nd}\sum_{i=1}^n\sum_{j=1}^d \left(\log x_{ij} - b_{i,2*}^{(t)} x_{ij}\mu_j^{-1}\right).
$$

Iterate until convergence, then we obtain the MLE $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$.

S1.2.3 LSGIGau-Ga$_d$ mean regression model

Assume that $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{ind}}{\sim} \text{LSGIGau-Ga}_d(\alpha, \boldsymbol{\mu}_i, \lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i, \boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observed data, we consider the following mean regression model

$$
\log E(X_{ij}) = \log\mu_{ij} = \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}_j \quad \text{or} \quad \mu_{ij} = \exp(\boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}_j), \quad i = 1, \ldots, n,\ j = 1, \ldots, d,
$$

where $\boldsymbol{z}_{(i)} = (z_{i1}, \ldots, z_{iq})^\top$ is a q-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta}_j = (\beta_{1j}, \ldots, \beta_{qj})^\top$ is the regression coefficients with $q \times d + 2 \leqslant n$. The log-likelihood function of parameter vector $\boldsymbol{\gamma} = \{\alpha, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_d, \lambda\}$ is

$$
\begin{aligned}
\ell_2(\boldsymbol{\gamma}|Y_{\text{obs}}) &= nd[\alpha\log\alpha - \log\Gamma(\alpha)] + \sum_{i=1}^n\sum_{j=1}^d [(\alpha-1)\log x_{ij} - \alpha\log\mu_{ij}]\\
&\quad - n\log 2 - n\log K_0(\lambda) + \sum_{i=1}^n \log\int_0^\infty h_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}\tau,
\end{aligned}
$$

where the function $h_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})$ is defined as

$$h_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) = \tau^{-d\alpha-1} \exp\left\{-\frac{1}{2}\left[a_0(\lambda)\tau + \left(b_0(\lambda) + 2\sum_{j=1}^{d} \frac{\alpha x_{ij}}{\mu_{ij}}\right)\cdot \frac{1}{\tau}\right]\right\}, \quad \tau > 0.$$

By defining the ndf

$$g_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) \triangleq \frac{h_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})}{\int_0^\infty h_2(s|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}s} = \mathrm{GIGau}\left(\tau\middle|a_0(\lambda), b_0(\lambda) + 2\textstyle\sum_{j=1}^{d} \alpha x_{ij}\mu_{ij}^{-1}, -d\alpha\right),$$

we construct the surrogate function as

$$\begin{aligned} Q_2(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)}) \;&=\; c_2^{(t)} + nd\left[\alpha\log\alpha - \log\Gamma(\alpha)\right] - nd\bar{d}_2^{(t)}\alpha \\ &\quad + \sum_{i=1}^{n}\sum_{j=1}^{d}\left(\alpha\log x_{ij} - \alpha\log\mu_{ij} - b_{i,2}^{(t)} x_{ij}\alpha\mu_{ij}^{-1}\right) \\ &\quad - n\log K_0(\lambda) - \frac{n\bar{a}_2^{(t)}}{2} a_0(\lambda) - \frac{n\bar{b}_2^{(t)}}{2} b_0(\lambda), \end{aligned}$$

where $c_2^{(t)}$ is a constant free from $\boldsymbol{\gamma}$, and for $i = 1, \ldots, n$,

$$\begin{aligned} a_{i,2}^{(t)} &= \int_0^\infty \tau\cdot g_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, & \bar{a}_2^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} a_{i,2}^{(t)}, \\ b_{i,2}^{(t)} &= \int_0^\infty \frac{1}{\tau}\cdot g_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, & \bar{b}_2^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} b_{i,2}^{(t)}, \\ d_{i,2}^{(t)} &= \int_0^\infty \log\tau\cdot g_2(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, & \bar{d}_2^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} d_{i,2}^{(t)}. \end{aligned}$$

Finally, we obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\gamma}}$ as

$$\begin{aligned} \alpha^{(t+1)} &= \mathrm{sol}\left\{C_2^{(t)} + \log\alpha - \psi(\alpha) = 0\right\}, \\ \boldsymbol{\beta}_j^{(t+1)} &= \left(\mathbf{Z}^\top\mathbf{W}_2^{(t)}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}_2^{(t)}\boldsymbol{r}_{j,2}^{(t)}, \quad j = 1, \ldots, d, \\ \lambda^{(t+1)} &= \arg\max_{\lambda>0} -\frac{n}{2}\left\{2\log K_0(\lambda) + \left[\frac{\bar{a}_2^{(t)}K_1(\lambda)}{K_0(\lambda)} + \frac{\bar{b}_2^{(t)}K_0(\lambda)}{K_1(\lambda)}\right]\lambda\right\}, \end{aligned}$$

where the constant

$$C_2^{(t)} \triangleq 1 - \bar{d}_2^{(t)} + \frac{1}{nd}\sum_{i=1}^{n}\sum_{j=1}^{d}\left(\log x_{ij} - \log\mu_{ij} - b_{i,2}^{(t)}x_{ij}\mu_{ij}^{-1}\right),$$

$\mathbf{Z} = (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})^\top$ is the matrix of covariates, $\mathbf{W}_2^{(t)} = \mathrm{diag}\{b_{1,2}^{(t)}, \ldots, b_{n,2}^{(t)}\}$ is the weighted matrix, $\boldsymbol{r}_{j,2}^{(t)} = (r_{1j,2}^{(t)}, \ldots, r_{nj,2}^{(t)})^\top$ with $r_{ij,2}^{(t)} = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j^{(t)} + (x_{ij}/\mu_{ij}^{(t)} - 1/b_{i,2}^{(t)})$ is the response vector for $i = 1, \ldots, n$.

S1.3 Multivariate lognormal mixture of Ga distribution

A positive r.v. Y is said to follow the *lognormal* (LN) distribution with location parameter $\mu \in \mathbb{R}$ and shape parameter $\sigma(> 0)$, denoted by $Y \sim \mathrm{LN}(\mu, \sigma^2)$, if its pdf is

$$\mathrm{LN}(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma y} \exp\left[-\frac{(\log y - \mu)^2}{2\sigma^2}\right], \quad y > 0.$$

We $E(Y) = \exp(\mu + 0.5\sigma^2)$ and $\mathrm{Var}(Y) = \exp(2\mu + \sigma^2)(\mathrm{e}^{\sigma^2} - 1)$. It is easy to show that if $Y \sim \mathrm{LN}(\mu, \sigma^2)$, then $c \cdot Y \sim \mathrm{LN}(\mu + \log c, \sigma^2)$, for any constant $c > 0$.

Let $\tau^* \sim \mathrm{LN}(\mu_\tau, \lambda)$, then we have

$$\mu^* \cdot \tau^* \stackrel{\mathrm{d}}{=} \mu^* \cdot \mathrm{LN}(\mu_\tau, \lambda) = \mu^* \exp\left(\mu_\tau + \frac{\lambda}{2}\right) \cdot \mathrm{LN}\left(-\frac{\lambda}{2}, \lambda\right) \stackrel{\mathrm{d}}{=} \mu \cdot \tau,$$

where $\mu = \mu^* \exp(\mu_\tau + \lambda/2)$ and $\tau \sim \mathrm{LN}(-\lambda/2, \lambda)$ with $E(\tau) = 1$, which can motivate the following definition.

S1.3.1 Definition and basic properties of the LN-Ga$_d$ distribution

Definition S1.3 (The multivariate LN-Ga distribution). Let $\tau \sim \mathrm{LN}(-\lambda/2, \lambda)$ and $X_j|\tau \stackrel{\mathrm{ind}}{\sim} \mathrm{Ga}(\alpha, \mu_j\tau)$, then the random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^\top$ is said to follow the *d-dimensional lognormal mixture of Ga* (LN-Ga$_d$) distribution, denoted by $X \sim \text{LN-Ga}_d(\alpha, \boldsymbol{\mu}, \lambda)$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_d)^\top$. ‖

Setting $f_\tau(\tau|\boldsymbol{\lambda}) = \mathrm{LN}(\tau| - \lambda/2, \lambda)$, we obtain pdf of $\boldsymbol{X} \sim \text{LN-Ga}_d(\alpha, \boldsymbol{\mu}, \lambda)$ as

$$\text{LN-Ga}_d(\boldsymbol{x}|\alpha, \boldsymbol{\mu}, \lambda) = \frac{1}{\sqrt{2\pi\lambda}} \left[\prod_{j=1}^d \frac{\alpha^\alpha x_j^{\alpha-1}}{\Gamma(\alpha)\mu_j^\alpha}\right] \int_0^\infty h_{3*}(\tau|\boldsymbol{x}, \boldsymbol{\theta})\,\mathrm{d}\tau.$$

where $\boldsymbol{\theta} \triangleq \{\alpha, \boldsymbol{\mu}, \lambda\}$, and

$$h_{3*}(\tau|\boldsymbol{x}, \boldsymbol{\theta}) = \tau^{-d\alpha-1} \exp\left[-\sum_{j=1}^d \frac{\alpha x_j}{\mu_j} \cdot \frac{1}{\tau} - \frac{(\log\tau + \lambda/2)^2}{2\lambda}\right], \quad \tau > 0. \tag{S1.4}$$

For $j, j' = 1, \ldots, d$ and $j' \neq j$, we obtain the expectation, variance and covariance as

$$E(X_j) = \mu_j, \quad \mathrm{Var}(X_j) = \mu_j^2 \left[\left(1 + \frac{1}{\alpha}\right) \mathrm{e}^{\lambda} - 1 \right] \quad \text{and} \quad \mathrm{Cov}(X_j, X_{j'}) = \mu_j \mu_{j'} (\mathrm{e}^{\lambda} - 1).$$

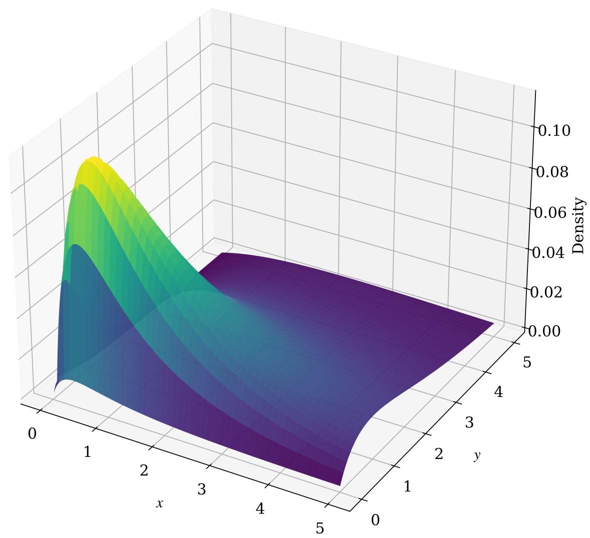


Figure 3: The density function of LN-Ga$_2$ distribution with $\alpha = 2$, $\boldsymbol{\mu} = (5, 3)^{\top}$, $\lambda = 0.5$.

S1.3.2 MLEs of parameters in LN-Ga$_d$ distribution

Let $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{LN-Ga}_d(\alpha, \boldsymbol{\mu}, \lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i\}_{i=1}^n$ denote the observed data, where $\boldsymbol{x}_i$ is the observation of $\boldsymbol{X}_i$, then, the log-likelihood function of $\boldsymbol{\theta}$ can be computed as

$$\begin{aligned} \ell_{3*}(\boldsymbol{\theta}|Y_{\text{obs}}) &= \sum_{i=1}^n \sum_{j=1}^d [\alpha \log \alpha + (\alpha - 1) \log x_{ij} - \log \Gamma(\alpha) - \alpha \log \mu_j] \\ &\quad - \frac{n}{2} \log(2\pi) - \frac{n}{2} \log \lambda + \sum_{i=1}^n \log \int_0^\infty h_{3*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}\tau, \end{aligned}$$

where $h_{3*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})$ is defined by (S1.4). By defining the ndf

$$g_{3*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta}) \triangleq \frac{h_{3*}(\tau|\boldsymbol{x}_i, \boldsymbol{\theta})}{\int_0^\infty h_{3*}(s|\boldsymbol{x}_i, \boldsymbol{\theta})\, \mathrm{d}s} \propto \tau^{-d\alpha - 1} \exp\left[-\sum_{j=1}^d \frac{\alpha x_{ij}}{\mu_j} \cdot \frac{1}{\tau} - \frac{(\log \tau + \lambda/2)^2}{2\lambda} \right],$$

we can construct the following surrogate function

$$
\begin{aligned}
Q_{3*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) &= c_{3*}^{(t)} + n\sum_{j=1}^{d}[\alpha\log\alpha - \log\Gamma(\alpha) - \alpha\log\mu_j] - nd\bar{d}_{3*}^{(t)}\alpha \\
&\quad + \sum_{i=1}^{n}\sum_{j=1}^{d}\left(\alpha\log x_{ij} - b_{i,3*}^{(t)}x_{ij}\alpha\mu_j^{-1}\right) - \frac{n}{8}\left(4\log\lambda + \lambda + 4\bar{s}_{3*}^{(t)}\lambda^{-1}\right),
\end{aligned}
$$

where $c_{3*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$ and for $1,\ldots,n$,

$$
\begin{aligned}
b_{i,3*}^{(t)} &= \int_0^\infty \frac{1}{\tau}\cdot g_{3*}(\tau|\boldsymbol{x}_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, & \bar{b}_{3*}^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} b_{i,3*}^{(t)}, \\
d_{i,3*}^{(t)} &= \int_0^\infty \log\tau\cdot g_{3*}(\tau|\boldsymbol{x}_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, & \bar{d}_{3*}^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} d_{i,3*}^{(t)}, \\
s_{i,3*}^{(t)} &= \int_0^\infty (\log\tau)^2\cdot g_{3*}(\tau|x_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, & \bar{s}_{3*}^{(t)} &\triangleq \frac{1}{n}\sum_{i=1}^{n} s_{i,3*}^{(t)}.
\end{aligned}
$$

We obtain the following iterative expressions:

$$
\begin{aligned}
&\alpha^{(t+1)} = \mathrm{sol}\left\{C_{3*}^{(t)} + \log\alpha - \psi(\alpha) = 0\right\}, \\
&\mu_j^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n} b_{i,3*}^{(t)}x_{ij}, \quad j=1,\ldots,d, \quad \lambda^{(t+1)} = 2\left(\sqrt{1+\bar{s}_{3*}^{(t)}} - 1\right),
\end{aligned}
$$

where the constant

$$
C_{3*}^{(t)} \triangleq 1 - \bar{d}_{3*}^{(t)} - \frac{1}{d}\sum_{j=1}^{d}\log\mu_j + \frac{1}{nd}\sum_{i=1}^{n}\sum_{j=1}^{d}\left(\log x_{ij} - b_{i,3*}^{(t)}x_{ij}\mu_j^{-1}\right).
$$

S1.3.3 LN-Ga$_d$ mean regression model

Assume that $\{\boldsymbol{X}_i\}_{i=1}^n \overset{\text{ind}}{\sim} \text{LN-Ga}_d(\alpha,\boldsymbol{\mu}_j,\lambda)$ and $Y_{\text{obs}} = \{\boldsymbol{x}_i,\boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observed data, we consider the following mean regression model

$$
\log E(X_{ij}) = \log\mu_{ij} = \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}_j \quad \text{or} \quad \mu_{ij} = \exp(\boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}_j), \quad i=1,\ldots,n,\ j=1,\ldots,d,
$$

where $\boldsymbol{z}_{(i)} = (z_{i1},\ldots,z_{iq})^\top$ is a q-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta}_j = (\beta_{1j},\ldots,\beta_{qj})^\top$ is the vector of regression coefficients with $q\times d+2 \leqslant n$. The log-likelihood function of parameter vector $\boldsymbol{\gamma} = \{\alpha, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_d, \lambda\}$ is

$$\begin{aligned}\ell_3(\boldsymbol{\gamma}|Y_{\rm obs}) &= nd[\alpha\log\alpha - \log\Gamma(\alpha)] + \sum_{i=1}^n\sum_{j=1}^d [(\alpha-1)\log x_{ij} - \alpha\log\mu_{ij}] \\ &\quad - \frac{n}{2}\log(2\pi) - \frac{n}{2}\log\lambda + \sum_{i=1}^n \log\int_0^\infty h_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}\tau,\end{aligned}$$

where $h_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) = \tau^{-d\alpha-1}\exp\{-\sum_{j=1}^d \alpha x_{ij}/(\mu_{ij}\tau) - (\log\tau + \lambda/2)^2/(2\lambda)\}$. By defining the ndf

$$g_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}) \triangleq \frac{h_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})}{\int_0^\infty h_3(s|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma})\,\mathrm{d}s}, \quad \tau > 0,$$

we construct the surrogate function as

$$\begin{aligned}Q_3(\boldsymbol{\gamma}|\boldsymbol{\gamma}^{(t)}) &= c_3^{(t)} - \frac{n}{8}\left(4\log\lambda + \lambda + 4\bar{s}_3^{(t)}\lambda^{-1}\right) - nd\bar{d}_3^{(t)}\alpha + nd\left[\alpha\log\alpha - \log\Gamma(\alpha)\right] \\ &\quad + \sum_{i=1}^n\sum_{j=1}^d \left(\alpha\log x_{ij} - \alpha\log\mu_{ij} - b_{i,3}^{(t)}x_{ij}\alpha\mu_{ij}^{-1}\right).\end{aligned}$$

where $c_3^{(t)}$ is a constant free from $\boldsymbol{\gamma}$, and for $i = 1, \ldots, n$,

$$b_{i,3}^{(t)} = \int_0^\infty \frac{1}{\tau}\cdot g_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \qquad \bar{b}_3^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n b_{i,3}^{(t)},$$

$$d_{i,3}^{(t)} = \int_0^\infty \log\tau\cdot g_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \qquad \bar{d}_3^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n d_{i,3}^{(t)},$$

$$s_{i,3}^{(t)} = \int_0^\infty (\log\tau)^2\cdot g_3(\tau|\boldsymbol{x}_i, \boldsymbol{z}_{(i)}, \boldsymbol{\gamma}^{(t)})\,\mathrm{d}\tau, \quad \bar{s}_3^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n s_{i,3}^{(t)}.$$

Finally, we obtain the $(t+1)$-th approximation of $\hat{\boldsymbol{\gamma}}$ as

$$\alpha^{(t+1)} = \mathrm{sol}\left\{C_3^{(t)} + \log\alpha - \psi(\alpha) = 0\right\},$$

$$\boldsymbol{\beta}_j^{(t+1)} = \left(\mathbf{Z}^\top\mathbf{W}_3^{(t)}\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\mathbf{W}_3^{(t)}\boldsymbol{r}_{j,3}^{(t)}, \quad j = 1, \ldots, d, \quad \lambda^{(t+1)} = 2\left(\sqrt{1 + \bar{s}_3^{(t)}} - 1\right),$$

where the constant

$$C_3^{(t)} \triangleq 1 - \bar{d}_3^{(t)} + \frac{1}{nd}\sum_{i=1}^n\sum_{j=1}^d \left(\log x_{ij} - \log\mu_{ij} - b_{i,3}^{(t)}x_{ij}\mu_{ij}^{-1}\right),$$

$\mathbf{Z} = (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})^\top$ is the matrix of covariates, $\mathbf{W}_3^{(t)} = \mathrm{diag}\{b_{1,3}^{(t)}, \ldots, b_{n,3}^{(t)}\}$ is the weighted matrix, $\boldsymbol{r}_{j,3}^{(t)} = (r_{1j,3}^{(t)}, \ldots, r_{nj,3}^{(t)})^\top$ with $r_{ij,3}^{(t)} = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}_j^{(t)} + (x_{ij}/\mu_{ij}^{(t)} - 1/b_{i,3}^{(t)})$ is the response vector for $i = 1, \ldots, n$.

S2. Some technical derivations and related distributions

S2.1 The root of $C + \log(s-1) + 1/(s-1) - \psi(s) = 0$ with $s > 1$ via a US algorithm

The aim of this appendix is to solve the root of the equation $g(s) = C + \log(s-1) + 1/(s-1) - \psi(s) = 0$ with $s > 1$ by employing a US algorithm. Since

$$\begin{aligned} g'(s) &= \frac{1}{s-1} - \frac{1}{(s-1)^2} - \psi'(s) > -\psi'(s) = -\sum_{m=0}^{\infty} \frac{1}{(m+s)^2} = -\frac{1}{s^2} - \sum_{m=1}^{\infty} \frac{1}{(m+s)^2} \\ &> -\frac{1}{s^2} - \sum_{m=1}^{\infty} \frac{1}{m^2} = -\frac{1}{s^2} - \frac{\pi^2}{6} \triangleq b(s), \end{aligned}$$

the US iteration for calculating $s^{(t+1)}$ is

$$\begin{aligned} s^{(t+1)} &= \mathrm{sol}\left\{ g(s^{(t)}) + \int_{s^{(t)}}^{s} b(z)\,\mathrm{d}z = 0,\ \forall\, s, s^{(t)} > 1 \right\} \\ &= \mathrm{sol}\left\{ g(s^{(t)}) + \frac{1}{s} - \frac{\pi^2}{6} s - \frac{1}{s^{(t)}} + \frac{\pi^2}{6} s^{(t)} = 0,\ \forall\, s, s^{(t)} > 1 \right\} \\ &= \mathrm{sol}\left\{ \frac{\pi^2}{6} s^2 - q^{(t)} s - 1 = 0,\ \forall\, s, s^{(t)} > 1 \right\} \\ &= \frac{q^{(t)} + \sqrt{q^{(t)2} + 2\pi^2/3}}{\pi^2/3}, \end{aligned}$$

where

$$q^{(t)} = C + \log(s^{(t)} - 1) + \frac{1}{s^{(t)} - 1} - \psi(s^{(t)}) + \frac{\pi^2 s^{(t)}}{6} - \frac{1}{s^{(t)}}.$$

S2.2 The generalized inverse Gaussian distribution

A positive r.v. Y is said to follow the *generalized inverse Gaussian* (GIGau) distribution (Jorgensen 2012) with parameters $a(>0)$, $b(>0)$ and $p(\in \mathbb{R})$, denoted by $Y \sim \mathrm{GIGau}(a, b, p)$, if its pdf is

$$\mathrm{GIGau}(y|a,b,p) = C_0 y^{p-1} \exp\left[-\frac{1}{2}\left(ay + \frac{b}{y}\right)\right], \quad y > 0,$$

where the normalizing constant C_0 is

$$C_0 = \left\{\int_0^\infty y^{p-1} \exp\left[-\frac{1}{2}\left(ay + \frac{b}{y}\right)\,\mathrm{d}y\right]\right\}^{-1} = \frac{(a/b)^{p/2}}{2K_p(\sqrt{ab})}$$

with $K_p(z)$ being the modified Bessel function of the second kind defined as

$$K_p(z) = \frac{1}{2}\int_0^\infty t^{p-1} \exp\left[-\frac{z}{2}\left(t + \frac{1}{t}\right)\right]\,\mathrm{d}t. \tag{S2.1}$$

For Bessel function $K_p(z)$, we have the following properties:

$$\begin{aligned} K_p(z) &= K_{-p}(z), \quad K_{p+1}(z) = \frac{2p}{z}K_p(z) + K_{p-1}(z) \quad \text{and} \\ K_{1/2}(z) &= \sqrt{\frac{\pi}{2z}}\exp(-z). \end{aligned}$$

And if $Y \sim \mathrm{GIGau}(a,b,p)$, we have the expectations of Y^k and $\log Y$ are as follows:

$$\begin{aligned} E(Y^k) &= \frac{K_{p+k}(\sqrt{ab})}{K_p(\sqrt{ab})}\left(\frac{b}{a}\right)^{k/2}, \quad k \in \mathbb{N}_+, \\ E(\log Y) &= \frac{1}{2}\log\left(\frac{b}{a}\right) + \frac{\partial}{\partial p}\log K_p(\sqrt{ab}). \end{aligned}$$

Specifically, when $a = 0$, $b = 0$ and $p = -0.5$, the GIGau distribution degenerates to the inverse gamma, gamma and inverse Gaussian distributions, respectively.

S2.3 The inverted beta distribution

A positive r.v. Y is said to follow the *inverted beta* (IBeta) distribution (Dubey 1970, Cordeiro & Lemonte 2012) with parameters $a\,(>0)$, $b\,(>0)$ and $p\,(>0)$, denoted by $Y \sim \mathrm{IBeta}(a,b,p)$, if its pdf is

$$\mathrm{IBeta}(y|a,b,p) = \frac{p^b y^{a-1}}{B(a,b)(p+y)^{a+b}}, \quad y > 0. \tag{S2.2}$$

The expectation, variance and mode of Y are given by

$$E(Y) = \frac{pa}{b-1},\ b > 1,\ \mathrm{Var}(Y) = \frac{p^2 a(a+b-1)}{(b-2)(b-1)^2},\ b > 2,\ \text{and } Y_{\mathrm{mod}} = \frac{p(a-1)}{b+1},\ a > 1,$$

respectively.

Remark S2.1 (Generating mechanism of inverted beta distribution). Let $X \sim \text{Beta}(a, b)$, define $Z \triangleq X/(1-X)$ and $Y \triangleq pZ = pX/(1-X)$ with $p > 0$, then Z follows the *beta prime* distribution (or beta distribution of the second kind) with pdf

$$\text{BetaPrime}(z|a, b) = \frac{z^{a-1}}{B(a, b)(1+z)^{a+b}}, \quad z > 0,$$

and $Y \sim \text{IBeta}(a, b, p)$. $\|$

S3. Supplementary tables in numerical experiments

S3.1 Accuracy of MLEs and reliability of asymptotic normal CIs

To assess the performance of our algorithms and methods, we conduct some numerical experiments by employing the *mean square error* (MSE) as the criterion to evaluate the accuracy of the MLE for each parameter and apply the *coverage rate* (CR) to confirm the reliability of CIs.

In our simulations, we consider different sample sizes $n = 300(300)900$ and $K = 10{,}000$ sampling replications. We conduct some numerical experiments with following case:

Case (A_2): Independently generating $\{\boldsymbol{X}_i^{(k)}\}_{i=1}^n \stackrel{\text{ind}}{\sim} \text{LSGIGau-Ga}_2(\alpha, \boldsymbol{\mu}_i, \lambda)$ with true values being set as $\alpha = 0.5$, $\boldsymbol{\beta}_1 = (1, 0, -0.5)^\top$, $\boldsymbol{\beta}_2 = (0, 2.5, -2)^\top$ and $\text{Var}(\tau) = 1$.

Given a combination of $\{n, k, \alpha, \boldsymbol{\mu}, \text{Var}(\tau)\}$ for Case (A_2), let $Y_{\text{obs}}^{(k)} = \{\boldsymbol{x}_i^{(k)}, \boldsymbol{z}_{(i)}^{(k)}\}_{i=1}^n$ denote the k-th observations with $k = 1, \ldots, K$, where $\boldsymbol{x}_i^{(k)}$ is a realization of $\boldsymbol{X}_i^{(k)}$ and the covariate $\boldsymbol{z}_{(i)}^{(k)} = (1, z_{1i}^{(k)}, z_{2i}^{(k)})^\top$ with

$$z_{1i}^{(k)} \stackrel{\text{iid}}{\sim} N(0, 1) \quad \text{and} \quad z_{12}^{(k)} \stackrel{\text{iid}}{\sim} \text{Poisson}(2), \quad \text{for } i = 1, \ldots, n.$$

We can compute the MLEs of parameters via N-EM algorithm, denoted by $\{\hat{\alpha}^{(k)}, \hat{\boldsymbol{\mu}}^{(k)}, \hat{\lambda}^{(k)}\}$. Next, we calculate the *average MLE* (A-MLE) with the corresponding MSE for each parameter based on the K repetitions. The interval estimation of parameters will be obtained by sandwich method, the CR of CIs will be presented in Table 1 together with A-MLE and MSE.

We found that with the increase of sample size n, the MLEs of parameters becomes more and more accurate and stable. Specifically, A-MLE is getting closer to the set true value, and MSE is getting smaller. For interval estimation, the CR is stable at about 95% with the sample size increasing, so a reliable interval estimation is obtained.

Table 1. Comparisons of A-MLE, MSE and CR for LSGIGau-Ga_2 distribution under various sample sizes

Sample size	Parameter	A-MLE	MSE	CR
300	α	0.5041	0.0009	0.9443
	β_{11}	0.9834	0.0379	0.9417
	β_{12}	−0.0013	0.0123	0.9354
	β_{13}	−0.5013	0.0060	0.9427
	β_{21}	−0.0120	0.0376	0.9409
	β_{22}	2.4994	0.0115	0.9491
	β_{23}	−2.0021	0.0060	0.9374
	$\text{Var}(\tau)$	0.9711	0.0553	0.9263
600	α	0.5020	0.0004	0.9516
	β_{11}	0.9930	0.0185	0.9474
	β_{12}	0.0006	0.0060	0.9437
	β_{13}	−0.5007	0.0029	0.9468
	β_{21}	−0.0054	0.0187	0.9452
	β_{22}	2.5008	0.0059	0.9486
	β_{23}	−2.0010	0.0030	0.9429
	$\text{Var}(\tau)$	0.9854	0.0281	0.9348
900	α	0.5014	0.0003	0.9469
	β_{11}	0.9952	0.0121	0.9520
	β_{12}	0.0004	0.0040	0.9379
	β_{13}	−0.5000	0.0019	0.9512
	β_{21}	−0.0042	0.0123	0.9459
	β_{22}	2.5016	0.0039	0.9485
	β_{23}	−2.0002	0.0019	0.9440
	$\text{Var}(\tau)$	0.9917	0.0184	0.9449

The convergence accuracy of MLE is 10^{-8}.

S3.2 Model comparisons

To assess the performance of the goodness-of-fit tests among the three specific distributions (i.e., IGa-Ga_d, LSGIGau-Ga_d and LN-Ga_d) in the Ge-Ga_d distribution family and other two new distributions (denoted by IGau-Ga$_d$ and RIGau-Ga$_d$, where IGau and RIGau denote *inverse Gaussian* and *reciprocal inverse Gaussian* distributions, respectively) in the Ge-Ga$_d$ distribution family, we adopt two evaluation criteria: The *Akaike information criterion* (AIC, Akaike 1974), and the *Bayesian information criterion* (BIC, Schwarz 1978).

In our numerical studies, the sample size, the number of sampling replications are respectively set to be $n = 500$ and $K = 1{,}000$. We set the parameters $\alpha = 2$, $\boldsymbol{\mu} = (2, 1.5, 0.8)^\top$ and λ in each of the five specific distributions (denoted by λ_{IGa}, λ_{LSGIGau}, λ_{LN}, λ_{IGau}, λ_{RIGau}, respectively) can be determined by assuming that $\text{Var}(\tau) = 0.5$.

For a given $\{n, k, \alpha, \mu, \text{Var}(\tau)\}$ with $k = 1, \ldots, K$, firstly, we independently generate the samples $\{\boldsymbol{X}_i^{(k)} = \boldsymbol{x}_i^{(k)}\}_{i=1}^n$ from the mixed distribution:

$$p_1 \cdot \text{IGa-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{IGa}}) + p_2 \cdot \text{LSGIGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{LSGIGau}}) + p_3 \cdot \text{LN-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{LN}})$$
$$+ p_4 \cdot \text{IGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{IGau}}) + p_5 \cdot \text{RIGau-Ga}_3(\alpha, \boldsymbol{\mu}, \lambda_{\text{RIGau}}).$$

where $p_s \geqslant 0$ for $s = 1, \ldots, 5$ with $\sum_{s=1}^5 p_s = 1$. We consider the following situations for selecting $\boldsymbol{p} = (p_1, \ldots, p_5)^\top$:

Case (B$_4$): Dominant one–component mixed distribution: One specific case of Ge-Ga$_3$ distribution has a larger weight p_s and the other two have smaller weights $p_{s'}$.

Based on the generated samples $\{\boldsymbol{x}_i^{(k)}\}_{i=1}^n$, we can compute the MLEs $\{\hat{\alpha}^{(k)}, \hat{\boldsymbol{\mu}}^{(k)}, \hat{\lambda}^{(k)}\}$ for the five specific distributions in the Ge-Ga$_3$ distribution family via N-EM algorithm, calculate five AIC values $\{\mathsf{AIC}_1^{(k)}, \ldots, \mathsf{AIC}_5^{(k)}\}$. Then we sort AICs to obtain the corresponding five ranks $\{\mathsf{Rank}_1^{(k)}, \ldots, \mathsf{Rank}_5^{(k)}\}$, where $\mathsf{Rank}_s^{(k)}$ represents the rank of $\mathsf{AIC}_s^{(k)}$ in $\{\mathsf{AIC}_1^{(k)}, \ldots, \mathsf{AIC}_5^{(k)}\}$. Finally, the average AICs (denoted by AIC) and the average ranks (denoted by Rank) are calculated by

$$\mathsf{AIC}_s = \frac{1}{K} \sum_{k=1}^K \mathsf{AIC}_s^{(k)} \quad \text{and} \quad \mathsf{Rank}_s = \frac{1}{K} \sum_{k=1}^K \mathsf{Rank}_s^{(k)}, \quad s = 1, \ldots, 5.$$

Tables 2 presents the AICs and Ranks of five fitted distributions. From the results in Tables 2, we found that the generated Ge-Ga$_3$ distribution provides the best fitting effect, which outstanding performance is significantly better than other distributions. In general, the

LSGIGau-Ga$_3$, LN-Ga$_3$ and IGau-Ga$_3$ distributions have good fitting effect in various mixed cases, means these distributions are robust to fit the complex data.

Table 2. Model comparisons based on $K = 1{,}000$ replications for Cases (B$_3$) and (B$_4$)

Mixed weight $\boldsymbol{p}$	Crit-erion	Fitted distribution				
		IGa-Ga$_3$	LSGIGau-Ga$_3$	LN-Ga$_3$	IGau-Ga$_3$	RIGau-Ga$_3$
$\frac{1}{10} \times (6,1,1,1,1)^\top$	AIC	**3512.3**	3513.7	**3512.5**	3512.8	3514.8
	Rank	3.22	2.67	**3.79**	**3.68**	1.64
$\frac{1}{10} \times (1,6,1,1,1)^\top$	AIC	3509.4	**3501.6**	**3501.5**	3501.6	3501.9
	Rank	1.36	**3.58**	**3.55**	3.38	3.13
$\frac{1}{10} \times (1,1,6,1,1)^\top$	AIC	3382.7	3381.7	**3380.9**	**3381.5**	3382.0
	Rank	2.43	2.89	**4.01**	**3.26**	2.42
$\frac{1}{10} \times (1,1,1,6,1)^\top$	AIC	3508.7	3502.8	**3502.6**	**3502.6**	3503.3
	Rank	1.55	3.48	**3.56**	**3.61**	2.80
$\frac{1}{10} \times (1,1,1,1,6)^\top$	AIC	3506.8	**3497.3**	**3497.2**	3497.6	3497.4
	Rank	1.22	**3.57**	**3.59**	3.21	3.41

The convergence accuracy of MLE is 10^{-8}.

Color Red has the best fitting performance, **bolding** is the second best.

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